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Reorienting psychological science

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Psychological phenomena occur across a wide range of scales, ranging from small, quick events of neurology and biology, to broader, more prolonged unfoldings typical of extended cultural practices. Although theories deployed by psychologists of different stripes have tended to incorporate these different scales, this is typically done in a manner that is implicit, and often unsystematic. That is, typical psychological research is conducted in a manner that is 'scale-blind'. In this article, I explore some of the historical and more recent recognition of this scale-blindness and place it in the context of recent work on the concept and implications of scale. I conclude by elucidating some of the important ways in which behaviour settings theory, and the researchers who developed it, are explicit and disciplined in their approach to scale, and how such scale-aware work promises practical value in improving scientific practice.

This article is part of the theme issue 'People, places, things, and communities: expanding behaviour settings theory in the twenty-first century'.

1. Introduction

The work of Roger Barker [1] and other researchers of the Midwest Psychological Field Station has a number of facets that set it apart from typical work in psychological science. In this article, I will argue that one of the crucial aspects of their research is the recognition that behaviour settings operate at a different scale to individual people and their streams of behaviour. Barker [1,2], in particular, argued for the need for a new kind of science, an eco-behavioural science, to recognize the implications of this scalar division, and to address questions about the operation of behaviour settings at a scale appropriate to the phenomena. Barker and the rest of the Midwest team (e.g. [3]) therefore demonstrate an explicit awareness of the issue of scale and the complexities that different scales introduce, which is uncommon in psychological science.

In what follows, I will briefly introduce the work of the Midwest team in order to illustrate this recognition of the different scales of behaviour. I will then draw on work from the ecological psychology of James J. Gibson, as well as that of Barker and his colleagues, to outline how perception and action are organized at different scales. Taking lessons from recent theoretical work on the concept of scale, I address how much of the history of psychological science is in fact marked by a blindness to scale—a tendency to overlook or dismiss scalar differences. I then turn to how scale is explicitly addressed by Barker and the Midwest team, and describe ways in which their work on behaviour settings can help psychological science to overcome its scale-blindness and to develop more robust and reliable research practices as a result.

2. Behaviour at scale

Recognizing that behavioural research was almost entirely conducted in the laboratory, and therefore that the scientific community had little information

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on behaviour in 'everyday life', Roger Barker and his colleagues organized a field research station in rural Kansas in the late 1940s. The town of Oskaloosa was given the pseudonym of 'Midwest', and the research team took time to integrate into the community to form working relationships and prepare the ground for extensive observations and recordings of the town's inhabitants in their natural settings. Thus was born the Midwest Psychological Field Station, whose work would continue for more than a quarter of a century.

Focusing initially on the activities of a small group of children who were observed from waking in the morning until going to bed at night, one of their primary expectations was to find patterns of stimuli (most likely social) that induced various particular behaviours in response.

In analysing their data, they were initially stumped, as even in the best cases only about half of all of the behaviours recorded had an identifiable antecedent stimulus. After significant effort in their analysis [1,4,5] the team came to recognize that, while behaviour was immensely variable, it was nevertheless highly structured. What is more, while particular actions by the children did not seem to directly follow specific stimulus presentations, it was in almost every case appropriate to, and meaningful within the situation in which the child was participating.

The Midwest team realized that higher-order structures of behavioural organization exist, and described them as 'behaviour settings'. A behaviour setting is a standing pattern of behaviour and its physical milieu. Settings typically have three principal aspects. They are often described almost entirely in terms of their physical characteristics—they are most frequently associated with particular places, which provide particular furniture and other appropriate resources. It is important to recognize, though, that the setting itself is not just the place, but includes the enactment of the pattern of behaviour too. A closed concert venue is just a big empty room. A concert, though, involves the venue, and its arrangement (a stage, perhaps seating, which will all face towards the stage), as well as people performing particular social roles—performers, audience members, maybe hospitality and security. Finally, these physical and behavioural aspects of settings are normally bounded by temporal ones. A setting comes into being, progresses through its typical pattern and then dissolves. Meetings are called to order then adjourned, a maths class finishes and then a geography class begins.

The slogan commonly used to describe the notion of behaviour settings is that 'People in school behave school; people in church behave church.' Through years of painstaking research, the Midwest team developed a detailed theory of behaviour settings and their operations [6]. For the purposes of the present article, however, the key point is the most broad one—human behaviour is organized not just on a moment-to-moment basis, but also at higher orders, on the broader scale. This shift in perspective is something that the Midwest team struggled to find, but once they did so, many fine-grained details started to make sense. It is this importance of appropriate orientation to a phenomenon—not just in details of precision, but at the broader remove—that I believe has profound implications for how we understand psychological science.

3. Turning to look

To perceive a thing, we must first be oriented towards it. Indeed, if we think about it, perceiving a thing *means* to orient our behaviour towards that thing to at least a coarse grain of resolution. The more closely we perceive it, the more finely oriented our behaviour can be. Orientation is the first broad sense of coordination by which our actions are organized with respect to the world around us.

It is in this sense that Gibson [7,8] has pointed out that our visual system is not simply a matter of the eyes and brain, but our whole bodies. Given that the eyes are at the front of our heads, and the interaction between them and the world around us is restricted (or directed) by the opacity of the skull, so our trunks, hips and legs play an enormous part in typical vision.

Orientation is not just a matter of the angle at which you encounter something, though. Distance matters too. Think of the way in which a person at an art gallery might step back or step forward in order to orient themselves to the work they are attending to, or perhaps more prosaically, the way in which someone entering middle age will move a text closer to or away from their face to be able to discern the letters clearly. Similar logic applies to other modalities.

In their examination of the complexity of the interaction between a cognitive agent and its environment, Erik Rietveld and Julian Kiverstein [9,10] address how the orientation of a person to their environment involves a complex of perception and readiness to act. Effective orientation at a broader scale of action both constrains and makes available fine-grained actions and sensitivities (see, in particular, Rietveld's discussion of a boxer positioning themselves in relation to an opponent [10]).

While it is a foundation of ecological psychology, the fundamental role of orientation in how animals perceive and act is something of a parable for scientific practice. To investigate something requires us as scientists to orient ourselves to it, and one of the most basic aspects of this orientation in the case of research is to address ourselves to the appropriate scale of the phenomenon.

4. Different scales do different things

Psychological science has long struggled with the question of scale, the result of which, as we will see, is a scientific endeavour that is somewhat disoriented. Before addressing that point, however, we should introduce some recent theoretical work that bears on the issue of scale, and specifically on the fact that phenomena at different scales are typically *not* just a matter of more of the same, but bigger in size or longer in duration.

Despite its clear importance to how we understand the world and how different aspects of the universe relate to one another, it is only recently that the concept of scale has been put to systematic examination. Two such recent investigations, Horton

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[11] and DiCaglio [12], both quickly reach the conclusion that differences of scale produce not just quantitative, but qualitative changes in events, dynamics and their implications.

It is perhaps trite to note that the move from the quantum to the classical domains in physics involves quite different kinds of phenomena, and concomitantly different appropriate methods of research and analysis. The tools of observation, recording and measurement appropriate to one scale become inappropriate, even obstacles, to another—the woods come to dominate where before we examined trees. What is an undifferentiable aggregate at one scale becomes a distinct but coherent phenomenon at another, with a different set of dynamics. The dynamics that govern the behaviour of the forest ecosystem are quite different from those of the individual trees, shrubs, fungi and wildlife that form it.

Both DiCaglio [12] and Horton [11] warn that because of these qualitative differences between the dynamics of phenomena at different scales, to orient towards a given scale necessarily implies orienting *away* from the others. We cannot coherently address two different scales of phenomena simultaneously. Indeed, it becomes a crucial methodological question as to how to mediate between theories of phenomena at different scales. Though we will return to the question of mediation later in the article, this issue gets philosophically complex very quickly, and would move us beyond the scope of present concerns. For our current purposes, we will simply acknowledge the inescapable difference of dynamics at different scales, a fact that is, as we have seen, at the heart of the insights and discoveries of Roger Barker and the rest of the team of the Midwest Psychological Research Station.

How is this rupture between scales of phenomena relevant to current practice in the psychological sciences? If we recognize the distinctness of happenings at different scales, we very quickly meet challenges to the way that much psychological research is conducted.

5. Scale-blindness in psychological research

Differing approaches to psychological science in part break down along scalar lines. Indeed, as we will see, the prospect of a unified psychological science is in part an illusion brought on by scale-blindness.

Though it is common practice to date the inception of psychology as an independent research endeavour to the founding of dedicated laboratories on either side of the Atlantic (Wundt's lab in Leipzig, James's in Harvard), the reality is that our present practices in many areas have histories that are independent of, and indeed precede, those events. The measurement and theorizing of inter-individual differences by Galton [13] were not inspired by the supposed founding figures, nor did they initially take place within departments of psychology. By 1957, Cronbach [14] would famously lament that two sciences of psychology existed and that they had little to do with one another. One derived from laboratory practice and was defined by the use of experiments; the other derived from the study of individual variation and gathered by Cronbach under the rubric of the use of correlations. It is both apt and telling that Cronbach identified these two bodies of literature with different methodologies, rather than different subject matter. Blind to the differences in phenomena that result from a shift in scale, the possibility that there might be different subject matters in question across these different efforts was not considered. Indeed, Cronbach's [14] concern was the lack of across these fields, which he considered to be a problem that was worth correcting.

In a follow-up study nearly twenty years later, Cronbach [15] was no more optimistic, however. Although experimental research had become sensitive to the interactions between experimental conditions and individual differences, the net result was, he feared, that 'we enter into a hall of mirrors that extends to infinity. However far we carry our analysis—to third order, or fifth order or any other—untested interactions of a still higher order can be envisioned' [15, p. 119]. Modern methods of mediation and moderation analysis may be designed to tame this proliferation, but as Borsboom *et al.* [16] and Yarkoni [17] address, the result is a game of theoretical and experimental whack-a-mole that undermines any hope of uncovering some hidden universal principles (see also Heino *et al.* [18] for a critical analysis of such linear methods).

Borsboom *et al.* [16] argue that psychology has not become any better integrated since Cronbach's pessimistic reviews, simply because these different approaches to psychology are really studying distinct phenomena. They cannot be integrated because they are not just different perspectives on the same thing, but they are, rather, methodological commitments that orient their researchers to quite different things. Borsboom *et al*. suggest that we move on, simply accepting as a negative conclusion the impossibility of simple integration—an acceptance that will allow us to better apportion resources and waste less time on dead ends. Though they do not explicitly address the question in terms of scale, their analysis separates two disciplines along scalar lines and argues (in a manner quite similar to [19]) that these separate strands do not bear an ambition towards integration. Cronbach's two disciplines of psychology really are just two disciplines of psychology.

Separately, Flis and van Eck's [20] bibliometric analysis of term use patterns in APA journal articles as indexed by PsycINFO appears at first glance to support this perspective. Examining five decades of publications, from 1950 to 1999, they map the use of 3000 key terms over that period (and individually within decades). The visualizations constructed via their analyses robustly support the idea of a separation between experimental and non-experimental psychologies. However, Flis & van Eck [20] do raise the question as to whether the 'correlational' psychologists are a coherent group, given that they include not only people involved in the questions of distributions of inter-individual differences, but also applied, individual person-focused domains such as clinical, educational and counselling psychology. Once again, here gathered under the straining umbrella of 'correlationists' are at least two scales of phenomena, this time not distinguishing populations from patterns of behaviour, but populations from individual persons in real-world settings. Though the bibliometric visualizations of Flis and van Eck confirm the broad separation of psychological research into separate lobes or hemispheres of research, they also suggest that scale has been encountered in more ways than just Cronbach's classic dichotomy.

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6. Psychology, a disoriented science

Contemporary psychological science addresses questions at a number of different scales. A brief reflection on the matter provides four possible such domains in what we might take to be typical mainstream research (though as the issue of these as different scales has seen little discussion, we might find questions arising about where to draw the lines between some):

- The biological: most frequently and explicitly neurological, but far from exclusively so.
- The behavioural: involving the measurement of individual behavioural episodes, the scale at which experimental *methods* are mostly frequently oriented.
- The personal: the level of individual human beings, the scale addressed most frequently by legal and civic considerations, particularly in the West; ironically probably the scale for which psychology has *least* developed theories or methods [21,22].
- The group: a broad scale spanning small-group samples to populations, the most frequent level of *analysis* of experimental research through the use of aggregationist statistics; also the level of both method and analysis of inter-individual research.

Orientation to a scale provides for a clear focus on a given phenomenon, but what seems to be unfortunately too-frequent practice within psychological science involves the conflation of scales and the undisciplined use of the same conceptual terms across these different domains without explicit recognition of the fact. Two examples will illustrate this. The first is that of cognitive psychological research on deductive reasoning (though the same pattern can be found in a wide variety of research fields).

A common form of research as typically conducted on reasoning invites participants to endorse conclusions on the basis of presented premises (e.g. [23,24]). Records are then taken as to the frequency of particular endorsements across a sample, and potentially other variables such as time to endorsement. We might also occasionally see such tasks carried out within an MRI machine, providing BOLD (blood oxygen level-dependent) signal observations during performance (e.g. [25]).

The ostensive purpose of such studies is to illuminate the processes by which inferences are produced by a generic, or standard cognitive system, presumably shared by all of the participants (perhaps as an evolutionary inheritance). To the extent that even this broad-brush outline is a fair description of the foundational logic of such research, we can see the extent to which this kind of psychological research is disoriented with respect to scale.

Let us take the four scales identified above. At the scale of groups of people, the data from a sample of participants (indeed, the larger the group the better) are collated, summarized and used to produce aggregate statistics for the purposes of analyses (more frequently through the use of some implementation of the general linear model). There are a number of assumptions underlying such use of aggregate statistics that are problematic here [26], but our particular focus is on the extent to which these characteristics of a group are taken to represent a process of inference, and in particular, the extent to which this process is the same thing that is broached by observations or conceptualizations that are applied at the other scales.

The same concept of 'inference' can be applied as the process represented by the group statistics, the process completed by a given person's standard cognitive endowment, the process underlying the pressing of a response button in a given instance of task performance and the process being captured by the recording of the BOLD signal in an fMRI study. That this reuse of the same concept is possible is indicative of the extent to which typical discourse in the psychological sciences conflates the dynamics of phenomena across scales; it is, indeed, oblivious to the distinctions between them.

I contend that this problem is not at all specific to the example of inference and reasoning here. Across the entire field of cognitive psychology, it is possible to see examples from decision making, to memory, to attention illustrating the same thing. Similar patterns are discernible in some areas of social psychology (particularly attitudinal research, for instance). It is in the field of individual differences research that this has perhaps come closest to being explicitly acknowledged, in the case of James Flynn's critique of the lack of discipline around the use of the term 'intelligence' [27].

Flynn describes three levels, rather than four, and does not make the explicit link to issues of scale (though as noted, there were no developed theories of scale available to him at the time). In reviewing the manner in which research on intelligence has been conducted, Flynn divides conceptualizations into three domains—the biological (the underlying neural processes involved in cognitive instantiation of intelligence), individual differences (intelligence as a dimension along which different people vary) and the social value (intelligence as virtue or ideal). Through examination of research in each field and the ways in which they relate to one another, but more importantly the ways in which they articulate and come apart when examined closely, Flynn diagnoses a confusion, a blindness to the independence of these frames of reference. The singular term 'intelligence' is deployed differently in each one, but then—precisely because it is a singular term—Flynn elides the unsupported conflation of one frame of reference with another, leading to what he refers to as 'paradoxes': situations where some notion of 'intelligence' is demonstrating seemingly inconsistent patterns of action.

These paradoxes dissolve once differences between concepts of intelligence are recognized and more carefully resolved, and the term is deployed and imposed in a more disciplined way. This conceptual confusion arises when the scalar orientation of the phenomena in which we are interested changes but our theoretical and empirical orientation does not.

There are a host of ways in which psychological research suffers from a lack of conceptual clarity [17,19,28]. What I mean to present here is one of the ways in which we can diagnose and begin to ameliorate that conceptual confusion, by taking the first steps to orient ourselves reliably.

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(a) How psychology has implicitly coped with scale

A scale-blind science will continue to encounter crucial differences that it cannot adequately conceptualize, theorize and investigate. Unable to address scalar differences explicitly, psychological science can only encounter and cope with them implicitly.

I suggest that to date, psychological scientists encounter and cope with scalar differences in three key ways. The first, and perhaps most common, is a reductionist philosophy of science that assumes a base (typically neurological) level of explanation from which the patterns of phenomena at all higher scales can *in principle* be derived, even if not in practice. There are two positive outcomes of a reductionist mindset for professional psychologists. The first is explanatory satisfaction when a neurological story is presented as underlying a given psychological phenomenon. Brain narratives *feel* scientific and compelling. The second, perhaps more operative outcome is that it collapses scalar differences. Behavioural-, personal- and group-level phenomena are conceived as simply aggregates, complex collections, of biological (or neurological) events. A reductionist perspective is, therefore, an implicit claim that scalar differences do not exist. While happenings at other scales might be useful descriptions of what is going on, they are nothing more than epiphenomena to the real action in the meat of body and brain.

A second, more reflective but also rather rare approach to scalar difference is given explicit articulation by Borsboom *et al.* [16]. These authors argue that phenomena at the scale of inter-individual differences in populations *supervene* on individual behaviour and its biological underpinnings, but that events at the higher scales are *multiply realizable*. There are many (perhaps a great many) different ways in which the same population-level phenomenon can be instantiated by things at the scale of people or particular behavioural systems. Such thinking leads Borsboom *et al.* [16] to express their pessimism regarding the integration of psychology as a coherent discipline, because the explanatory asymmetry between intra-individual processes and inter-individual differences means that the two disciplines of psychology are ultimately just asking questions about different things.

Given the lack of simple reducibility across scales argued for in depth by DiCaglio [12] and Horton [11], which I have endorsed here, and in keeping with the analysis offered by Borsboom *et al.* [16], it would seem that we should just allow the two different aspects of psychology to de-couple from one another and be done with it. This view, however, is made problematic by the third, implicit and much more problematic manner in which psychological science has tended to encounter scalar differences. That is as context effects.

By 'context effects' here, I refer to the broad, varied kinds of phenomena across different sub-fields of psychology that result in the complex web of interactions bemoaned by Cronbach [15]. When people do things, what matters is not simply the specific stimuli or movements involved, but the whole situation. Repeatedly, we find that if we observe and measure specific behaviours (including similar tasks or experimental conditions) in different circumstances, we get different results (Cronbach [15] provides some nice examples—e.g. [29,30]). What context effects tell us is that interaction between scales is not simply the reductionist means by which we have 'bottom-up' influence on happenings, nor are the distinct dynamics of higher scales entirely independent of the lower, such that sciences of different scales can afford to be wholly ignorant of what goes on between them. Instead, complex feedback relations exist between scales that cannot be ignored.

7. Contextual causality

Alicia Juarrero [31] has examined these inter-scale relationships in detail, arguing that our increasing understanding of them, through the use of tools such as dynamical systems theory and complexity theory, transforms our understanding of causation. In particular, she argues that the exclusive fixation on efficient causation must be relaxed in order for us to better understand how the universe is organized.

Efficient causation is the standard conception deployed in science, in which one object at a given scale induces a change in another at the same scale through physical contact. Juarrero [32] had previously argued about the problematic limitations of this overly narrow view of causation with regard to human action. In her recent work, *Context changes everything* [31], she articulates a concept of constraint-as-causation that addresses not within-scale interactions, but situations in which phenomena at one scale are affected not by contact with another object, but through constraints arising by being embedded in phenomena occurring at another scale.

In our currently scale-blind state, we cannot properly orient ourselves towards these phenomena. Extant, dominant methods do not bring them into view. The result is a literature replete with struggles to reliably produce phenomena, and no wonder. Typical (though by no means all) psychological research methods are a mish-mash of scales, with no clear theory as to how they relate to one another. We record specific behavioural events by individual participants but analyse aggregated, averaged responses [26,33,34]. We attempt to isolate elementary behavioural or cognitive processes, but do so by trying to produce them in the laboratory, away from the kinds of contexts—pressures, motivations, norms and other forces—in which such events are embedded in everyday life, leaving a literature replete with the kinds of 'context effects' lamented by Cronbach [15] and little improved on since ([16]; see also [17] for an importantly related discussion on the impossibility of completed linear models).

There is a lot of work to be done in order to orient our investigations effectively and to allow us to observe, measure and theorize what scales there are, and how they interact with one another, constraining and mediating one another.

The first step is to make ourselves aware of the possibility of these differences and to begin being explicit about them. The second is to expand our set of methods to allow other scales of phenomena to come into view. And as it happens, it is in these efforts that we find Roger Barker, and his colleagues at the Midwest Psychological Field Station, have already begun the work.

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8. Behaviour settings demarcate context

A notable exception to the claim that psychological scientists have not explicitly engaged with the issues and implications of scale is that of ecological psychology, in both the form derived from the work of James and Eleanor Gibson (and those who have followed them) and that of Roger Barker and the rest of the team at the Midwest Psychological Field Station.

Ecological psychology is defined by the recognition of the *ecological scale*, the scale at which an embodied agent engages with and interacts with the world around them. For Gibson [35], the ecological scale is defined relative to the body of the active perceiver or agent. The body of the animal means that some aspects of the world matter are relevant to and appropriate to the animal, while others are not. The ecological scale is the scale at which the world is *meaningful* for the agent in question [36]. With regard to the four scales enumerated above, Gibson's ecological scale is at the level of the individual (whole, embodied) person, precisely the scale least addressed by more mainstream forms of psychological research. Crucially, however, Gibson [35] emphasized that the environment (and by extension, the actions we take as we engage with it) is *nested*, which is to say that it has structure at a variety of scales. The vast majority of ecological research that has been undertaken was done at the scale of the single agent—a whole person getting something done at the level of the task or behaviour episode. (A very recent work by Damian Kelty-Stephen [37] raises important questions about the coherence of the 'ecological scale' as discussed by Gibsonian researchers. Kelty–Stephen draws conclusions about the need for a more careful consideration of scales very similar to some of those indicated here, though following a somewhat different path.)

Somewhat confusingly, the ecological scales of the Gibsons and the Midwest team are actually a little different from one another, but because they are scale aware, both approaches allow for this difference. Heft ([5,38,39]; see also [40] in this issue) notes that behaviour settings are group-level phenomena. Barker [1,2] and his colleagues (e.g. [3]) describe behaviour settings as an 'eco-behavioural' phenomenon. Making sense of the behavioural data from their naturalistic recordings at the scale of the individual actor and the immediate 'stimuli' or 'inputs' was difficult to impossible until a distinct scale of the setting, and its shaping influence on the behaviour of setting inhabitants, became apparent [1]. It is a consistent observation of myself and others working with the concept of behaviour settings that, once described, the phenomenon is almost impossible to deny. Behaviour settings are *obvious*. Indeed, this is perhaps one of the things that has ensured their wallowing in obscurity as a domain of research (along with the structuring of professional incentives for research; see [41] and Heft [40] in this issue). Their intuitiveness makes the curious keen to move beyond them, to seek explanatory satisfaction, usually at a (reductionist) finer grain of resolution (the individual in the setting).

Juarrero's [31] argument, however, suggests that we should be cautious in moving on so quickly. Barker and others' claim that behaviour settings are a phenomenon on a scale of their own suggests that they have their own dynamics and that their influence on the behaviours of their inhabitants is not a simple sum of their interacting components. The molar level of the setting is not a simple additive sum of the individual behaviours of its inhabitants.

Entirely consistent with Juarrero's account, in recent work, Heras-Escribano & Raja [42] have explicitly described behaviour settings as providing what they call 'enabling constraints'. Enabling constraints are constraints that, as a result of the limitations they impose, make some things either possible at all, or more likely. They describe means by which this mode of analysis can be made continuous with modelling practice more generally in Gibsonian ecological psychology more generally, which tends to be task-focused.

At the time that the Midwest team were undertaking and publishing their work, the claim about the eco-behavioural scale was empirically driven, but not finely testable. Developments in dynamical systems theory since, however, provide some tantalizing suggestions (see Heft [40] in this issue for a related point).

Specifically, an extensively evidenced set of findings indicate that well-functioning, integrated phenomena demonstrate particular kinds of distributions in their variability. Chemero [43] provides a very useful summary and introduction to this work [44], which has been repeatedly and extensively replicated across a wide range of disciplines. Phenomena that self-organize into what we can consider coherent wholes reliably demonstrate particular distributions of variability. Such distributions indicate that changes at one time have a prolonged influence over the course of a systems' operation such that later variations are related to earlier ones—what is termed 'autocorrelation'. The presence or absence of this autocorrelation over multiple timescales has been shown to distinguish well-functioning from unhealthy hearts [45], for instance. This temporal interdependence is also characteristic of reaction times for laboratory cognitive tasks [46] and a bewildering array of other naturally occurring systems [47]. While certain simple random noise generator systems can demonstrate autocorrelation, which results in a simple fractal character to the variability of such systems, natural systems tend to have what we might simplify as multiply layered autocorrelation, or what is termed *multifractality*.

Chemero [43] reasonably contends that the ubiquity and reliability of multifractality in naturally occurring systems mean that it can be used diagnostically for the existence of an interaction-dominant system—or as I have more bluntly put it, as an indication that the system is a coherent whole rather than a collection of individual components. Barker's claim that behaviour settings are a distinct scale of phenomenon therefore suggests a newly testable empirical hypothesis that variability within properly operating settings will display such multifractality. Speculatively, as higher order phenomena, it may be that such variation will be found between existing multifractal processes within a setting. Perhaps conversational turn-taking and limb movements during setting-specific actions may show autocorrelation not just within each action, but between them too.

The work of the Midwest team offers more than this one speculative potential, however. It is becoming clear that psychologists will have to become increasingly comfortable with such dynamical systems analyses, but even within more mainstream practices and higher-level discourse on behaviour and experience, behaviour settings theory provides value.

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9. What behaviour settings can do for us

(a) Being scale aware

I have argued that contemporary mainstream psychological science is disoriented, oscillating between scales across observation, conception and analysis, with little discipline. The result is too frequent misapplications of theoretical concepts [27] or incoherent frames of reference in the formation of research questions and the interpretation of results [34].

Dynamical approaches to psychology tend to be scale aware. Where a network or complex systems perspective is adopted, there is an explicit recognition that phenomena occur across multiple scales (e.g. [48–50]).

However, theorization of those scalar differences, and a systematic, careful examination of their interaction, are less well developed even where these dynamic techniques are quite advanced. DiCaglio [12, p. 11], for instance, laments this limited character of scalar theory, noting that even in a more accessible overview source such as Geoffrey West's '*Scale: the universal laws of life, growth, and death in organisms, cities, and companies*' [51], there is recognition of some of the characteristics of scalar differences but no analysis providing a clear framework for thinking systematically about how scale comes to have the impact that it does. This dynamical work provides vital insights into the importance of self-organization of emergent properties between the scale of the whole system and that of its component parts. But there remains work to be done to identify theoretical constructs that enable effective communication and coordination between scientists working at those different scales as well as theorization of different basic or consistent scales of psychological phenomena, as the work of the Midwest team suggests do in fact exist.

The first step to addressing either the scale-blindness of mainstream psychology or the limited articulations of scale in more dynamic fields is simple to say, but tricky to manage: to be explicit about the scale at which we are working at all times. When we have a consistent axis of orientation, changing perspective in itself is not a problem or an error. The difficulties arise when the perspective has changed implicitly, and our actions thus become inappropriate—it is a bit like acting as though we do not need to adapt our arm movements in picking up a cup depending on whether or not we are facing the table. With regard to scale, our professional vocabulary has been very limited. Behaviour settings theory provides a good start.

Of the four rough scales that we can quickly identify on reflection, perhaps the most vague is that of the group. Without being properly disciplined, the notion of a 'group' scale extends across the entire range from a dyadic interaction to population dynamics. Behaviour settings theory provides a systematic vocabulary which identifies a consistent domain that is relevant to the organization of behaviour at the molar level. It describes a meaningful level that is independent of the individual people involved. In doing so, it makes deliberate and careful scale awareness possible in a way that allows us to think clearly and carefully about context and the production of behaviour.

(b) Context and external validity

In a sense, context is at the core of the replication crisis in behavioural science (something Cronbach foresaw [15]). In circumstances where we do not have a clear theory of context, or how the organization and production of behaviour are affected by the broader domain in which they occur, then evaluating the validity of any laboratory finding is a fraught project. Different psychological phenomena that present in the same behaviour, or the same underlying psychological dynamics that results in different behaviours, cannot be recognized because there is no surrounding context to support their systematic and disciplined categorization.

In the concept of context, the issues of external and ecological validity are intertwined. The extent to which we can be confident that findings from laboratories represent dynamics and events in the everyday world is the extent to which the key relations of the laboratory activity reflect activity outside of the lab. Brunswik's [52] initial characterization of ecological validity—that the probability distributions of particular stimuli in the lab should match (as closely as possible) those in the field—was essentially atheoretical. With behaviour settings, we can move beyond such simple notions of distributions, to recognize that real-world behaviour occurs in behaviour settings that have particular kinds of structure—roles of inhabitants, relations between behaviours and furnishings and more. Using an understanding of behaviour settings, we can derive a more sharply defined conception of external validity as circumstances in the laboratory that *maintain the key relations* between operative aspects of a given behaviour setting. It seems clear that where the key facets of behaviour settings have been preserved in the controlled environment of the laboratory, then our confidence in the external validity of the observations in such circumstances is increased. We can deliberately and systematically investigate the context of what we are trying to understand, which is something that can only be a boon both to the formation of valid research questions, and to the likely reliability of the conclusions that we draw in answering these questions on the bases of the actions to answers to those questions.

(c) A new domain of research

Barker and the Midwest team have done a great service in identifying a naturally occurring phenomenon that is both powerfully structuring behaviour and is readily identifiable. In the context of a revised conception of causality across scales presented by Juarrero [31], building on the work of Van Orden *et al*. [46] and others (see [43] for an overview), we find ourselves with a new domain of research opening up. Phenomena at different scales do not simply interact in the same way as objects at the same scale do. Both Horton [11] and DiCaglio [12] emphasize that the interaction between different scales is one that relies on particular kinds of mediation. It is not coincidental that the behaviour settings research group make much of the

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concept of the psychological medium, and the relationship between the setting and the behaviour of its inhabitants as one dependent on our understanding of what a behavioural medium might be [1, pp. 159–167; 6, pp. 169–175]. Making ourselves scale aware, therefore, opens up a new domain of research, exploring the structures and processes of this inter-scale mediation. While behaviour settings work *per se* is to be undertaken at the scale of the setting and has nothing specific to say about individual persons as individuals [3] the processes by which they come to have such powers of behavioural organization at the individual level—which we might consider behavioural context studies—will play a crucial role in understanding the traditional phenomena of the psychological sciences.

10. Conclusion

I have argued in this article that psychological science has a scale problem. Problems in psychological research—such as the paradoxes in intelligence research and conceptual slippage in cognitive psychology—probably underlie some of the difficulties with replication and are a product of a scale-blindness, an undisciplined shifting of perspectives from one scale to another, applying the same concepts in these domains without regard for the vital ways in which different scales work differently [11,12].

Finding our way out of this confusion will require us to be more explicit about the scale at which our research is directed, and to be systematic and deliberate in the way that we apply concepts appropriate to that scale. Essential to this change of practice is having a clear description of relevant scales. While some are relatively straightforward, e.g. that of individual behaviour episodes typically recorded in experimental research, the scale of the behaviour setting is one that clearly has a powerful organizing influence on behaviour but has not been explicitly recognized outside of the work of the original Midwest team. The inter-scale, dynamically self-organizing character of behaviour is becoming increasingly recognized. Given how it makes long-trenchant problems such as context and behavioural variability more tractable, the role of behaviour settings in orienting and anchoring the observations and measurements in our scientific practice will become ever more valuable.

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